

Quantum Radar for Battleship Game using Hybrid Quantum Algorithms (QFF25004)

Abstract & Introduction

Abstract

- The project demonstrates a **quantum radar system** for the Battleship game.
- Uses **Quantum Illumination** and **Grover's Algorithm** to detect ships faster than classical methods.
- Combines **entanglement**, **superposition**, and **amplitude amplification** for high accuracy under noise.
- Hybrid model improves detection efficiency and reduces computational cost.

Problem Statement

- Classical search $\rightarrow O(N^2)$ time complexity for an $N \times N$ grid.
- Cannot handle **noisy or probabilistic detection**.
- Objective: Develop **quantum-enhanced detection** with faster and more accurate identification of hidden ships.

Proposed Solution Overview

- **Quantum Illumination:** Detects *presence* of ships using photon correlations.
- **Grover's Algorithm:** Identifies *exact* ship positions efficiently.
- **Hybrid System:** Illumination reduces search space $\rightarrow$ Grover refines results.

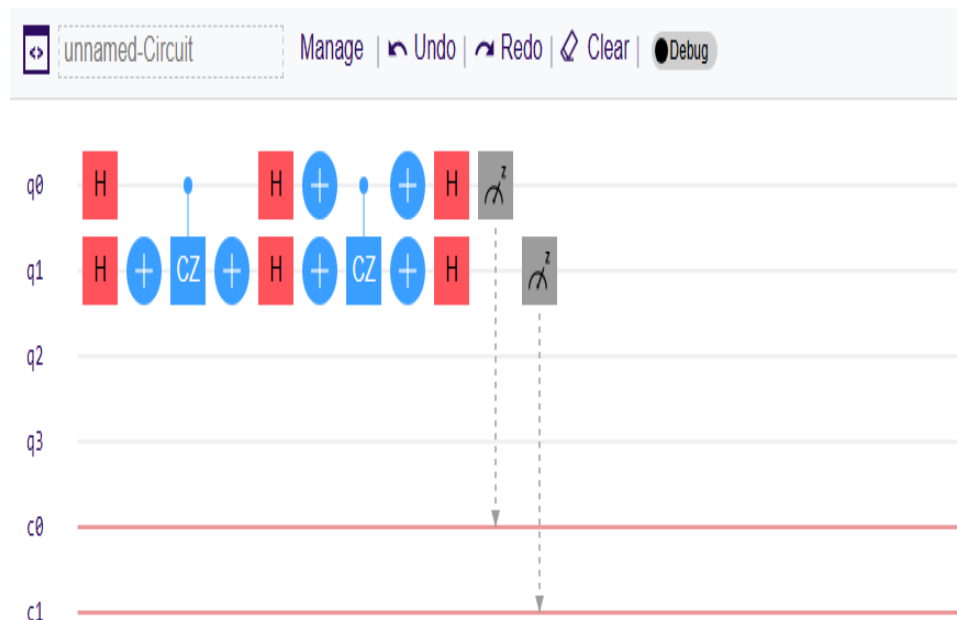
Theoretical Background & Optimized Circuit Design

Quantum Concepts Used

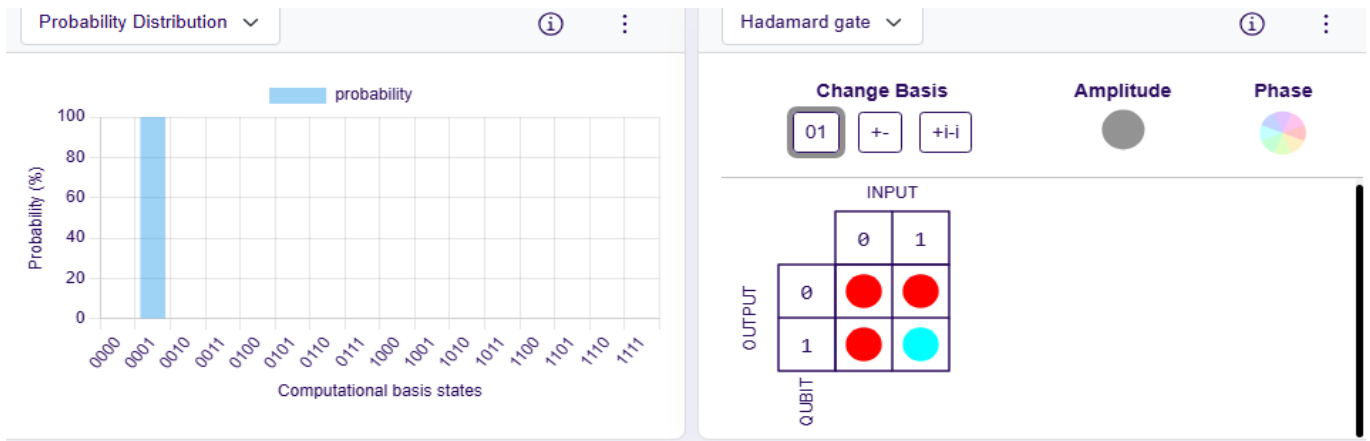
- **Superposition** $\rightarrow$ scans multiple grid cells simultaneously.
- **Entanglement** $\rightarrow$ links photon states (signal & idler).
- **Amplitude Amplification** $\rightarrow$ boosts target probabilities via Grover iterations.

Optimized Circuit Design

- **2-Qubit Bell Pair** per cell (signal + idler).
- Operations:
 - H $\rightarrow$ superposition
 - CZ $\rightarrow$ entanglement
 - Conditional Z $\rightarrow$ ship presence phase flip
 - Measurement $\rightarrow$ ship probability outcome



- **Circuit depth minimized** using only three main gates per pair.
- Output represented with **histogram** of correlation probabilities ($|00\rangle$, $|11\rangle$ states).



Methodology & Implementation

Step 1: Quantum Illumination

- Generate entangled photon pairs.
- Send **signal photon** toward the grid; **idler photon** retained.
- Measure correlations $\rightarrow$ compute detection probability $P(i,j)$.
- Select cells where $P(i,j) > \text{threshold}$.
- Output: *Detection probability map* (heatmap visualization).

Include Visual: Heatmap (probabilistic detection map).

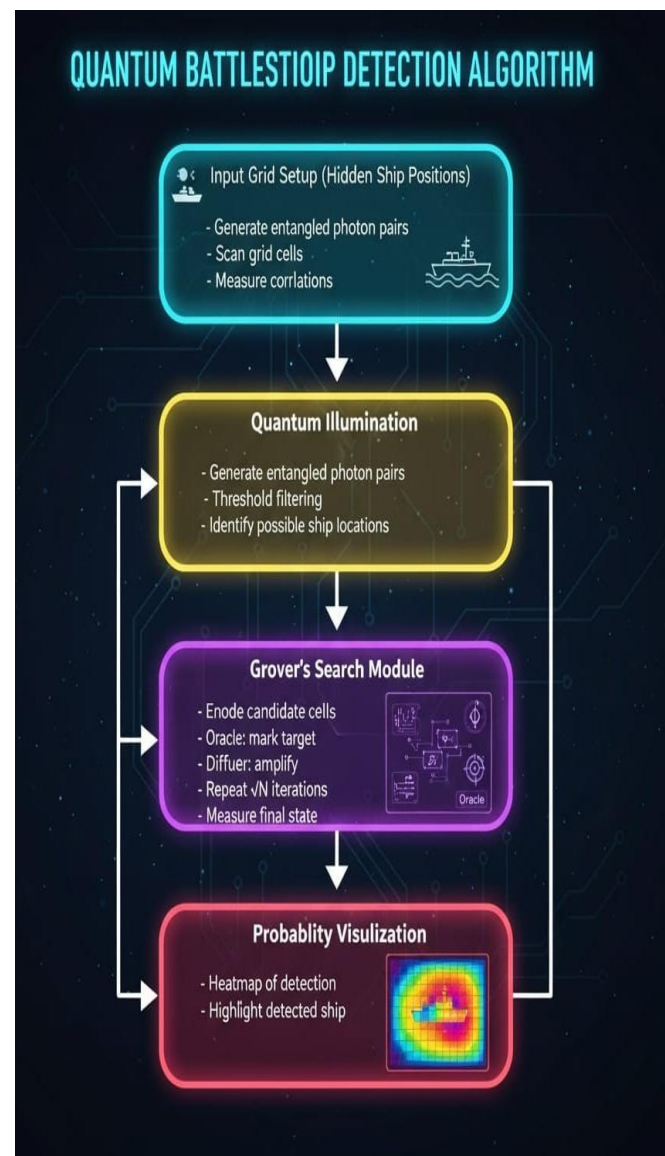
Step 2: Grover's Quantum Search

- Encode candidate cells from Step 1.
- Oracle marks true ship states.
- Diffuser amplifies correct amplitudes.
- After $\sqrt{N}$ iterations $\rightarrow$ measure quantum register.
- Output: *Exact ship locations*.

Include Visual: Pie chart comparing quantum vs classical efficiency.

Step 3: Visualization Module

- Combine detection map and Grover output.
- Overlay actual vs detected ships.
- Include **heatmaps**, **histograms**, and **probability bar graphs**.

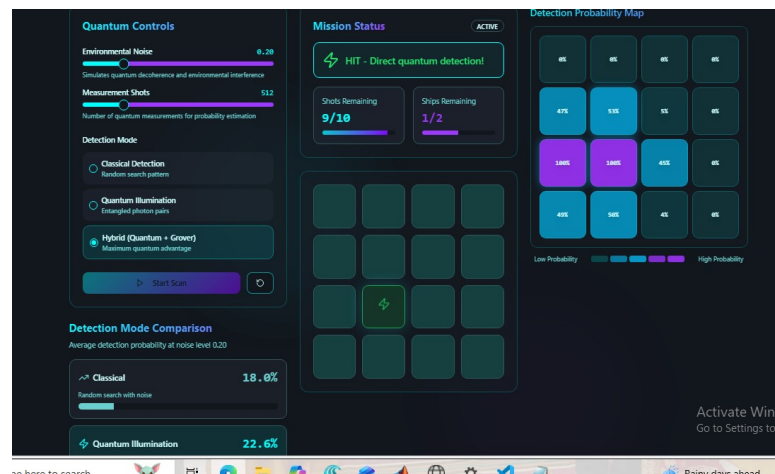


Demo Implementation (Website + Code Execution)

- **Frontend:**
 - Built with **React.js** + **Plotly.js** for interactive visualization.
 - Displays **detection heatmap**, **Grover search results**, and **probability bar graphs**.
 - User can choose grid size and run quantum simulation live.
- **Backend:**
 - **Python (Flask)** server handles logic and computation.
 - Integrated **Qiskit AerSimulator** for local quantum circuit execution.
 - Code executes entangled photon creation, correlation measurement, and Grover iterations.

Demo Workflow:

- User selects grid (e.g., 4×4).
- “Run Simulation” triggers backend quantum code.
- System displays:
 - **Detection Probability Map (Heatmap)**
 - **Measurement Histogram (|00⟩, |11⟩)**
 - **Grover Result Highlight (Exact Ship Position)**
- **Hosted Prototype:**
<https://lovable.dev/projects/1216444d-9e59-46ac-a088-b13f1f0c9e1b>
- web interface (Heatmap + Histogram view).



Results, Optimization & Evidence of Quantum Learning

- **Simulation Setup**
- Tools: **Python, Qiskit AerSimulator, Matplotlib**
- Environment: **4×4 or 5×5 grid**, 512 shots.
- Outputs generated for both **quantum** and **classical** cases.

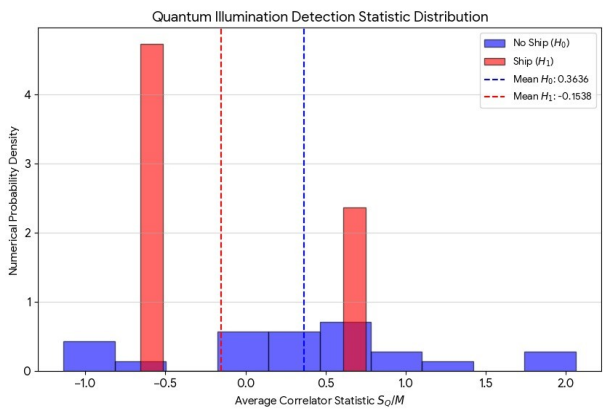
Results

- Quantum heatmaps → show high probability regions.
- Histograms → correlation peaks in $|00\rangle$, $|11\rangle$.

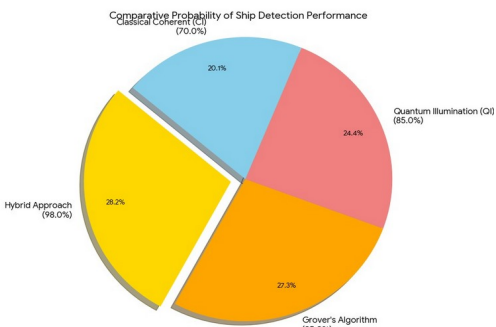
- Grover’s search → detects exact ship cells within 2–3 iterations.
- Classical radar → requires ~16 iterations for same grid.

Evidence of Quantum Learning

- **Entanglement effects:** correlated signal–idler outcomes.
- **Noise resilience:** detection persists under weak signals.
- **Amplitude amplification:** Grover enhances correct candidate probabilities.
- **Circuit output histograms:** prove quantum superposition and measurement logic.
- **Include Figures:**



Histogram (measurement outcomes)



pie chart(comparison)

Feasibility, Advantages & Conclusion

Feasibility and Comparison

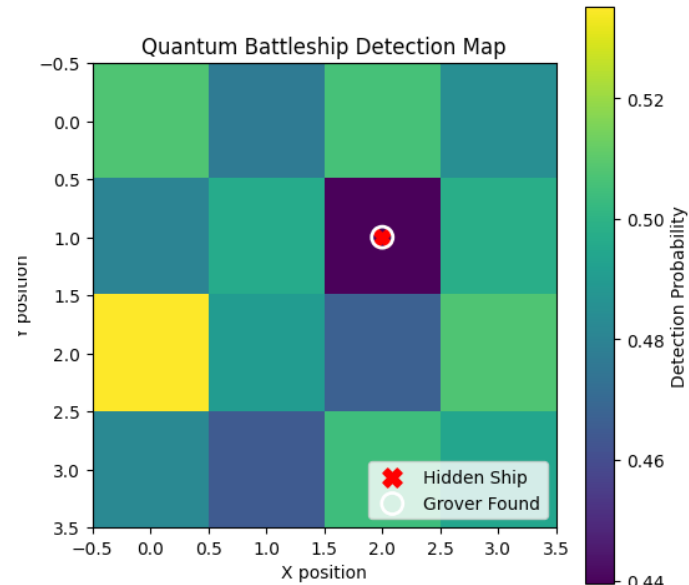
Aspect	Classical Radar	Quantum Battleship
Detection	Based on reflected amplitude	Based on entangled photon correlations
Noise Tolerance	Low	High
Search Complexity	$O(N^2)$	$O(N)$
Accuracy	Probabilistic	Near-deterministic via amplitude amplification

Advantages

- **Time-efficient:** Fewer iterations ($\sqrt{N}$).
- **Scalable:** Works for larger grids.
- **Noise-tolerant:** Quantum Illumination effective in low-signal scenarios.
- **Hybrid quantum-classical:** Merges detection and localization.

Conclusion

- Successfully demonstrates **quantum-assisted detection** for Battleship.
- Quantum Illumination + Grover hybrid achieves both **speed** and **accuracy**.
- Supports visualization via **optimized circuits**, **heatmaps**, and **histograms**.
- Future scope: Implement on IBM Quantum hardware for real-world validation.



Future Scope

- **Real Quantum Hardware Execution:** Run on IBM Quantum processors to validate real-time results.
- **Multi-Ship Detection:** Extend hybrid algorithm for simultaneous multi-target localization.
- **Dynamic Environments:** Apply adaptive noise modeling for moving ship targets.
- **Quantum-Classical Fusion:** Integrate AI (e.g., reinforcement learning) for adaptive oracle creation.
- **Hardware Optimization:** Reduce decoherence via gate scheduling & error correction.
- **Educational Deployment:** Use the demo for quantum learning in universities and hackathons.
- **Integration with Quantum Networks:** Explore entangled communication for distributed radar systems.

References & Citations

- M. Lloyd, "Quantum illumination: Detecting the presence of an object using entanglement," *Science*, vol. 321, pp. 1463-1465, 2008.
- L. K. Grover, "A fast quantum mechanical algorithm for database search," *Proceedings of the 28th Annual ACM Symposium on Theory of Computing*, 1996.
- IBM Quantum Documentation — <https://quantum.ibm.com> OpenAI, ChatGPT (Model GPT-5), assisted in generating technical documentation, optimization ideas, and simulation planning for this project.
- Google DeepMind, Gemini AI system, used for comparative reference in AI-assisted document synthesis and visualization methodology.
- YouTube Channel — "RGUKT Srikakulam Official", which hosts lectures and workshops organised by IBM Qiskit Fall Fest 2025 at RGUKT Srikakulam:

[https://youtube.com/@rgukt_srikakulam_official?
si=x3qbcfTMubU0yRXg](https://youtube.com/@rgukt_srikakulam_official?si=x3qbcfTMubU0yRXg)